

DRUG SAFETY SIGNAL DETECTOR

& Regulatory Submission Readiness Checker

From Safety Signal Detection to Regulatory Submission Readiness

Two Bottlenecks, One Broken Handoff

Pharmacovigilance teams and regulatory affairs teams both drown in manual review — and today they work apart.



Adverse-event volume

Huge volume of adverse-event reports makes manual safety analysis difficult.



Prioritization is hard

Potential drug-event associations are difficult to prioritize by hand.



Dense CTD dossiers

Pharmaceutical dossiers contain large amounts of CTD documentation.



Manual gap-finding

Finding missing regulatory documentation manually is time-consuming.

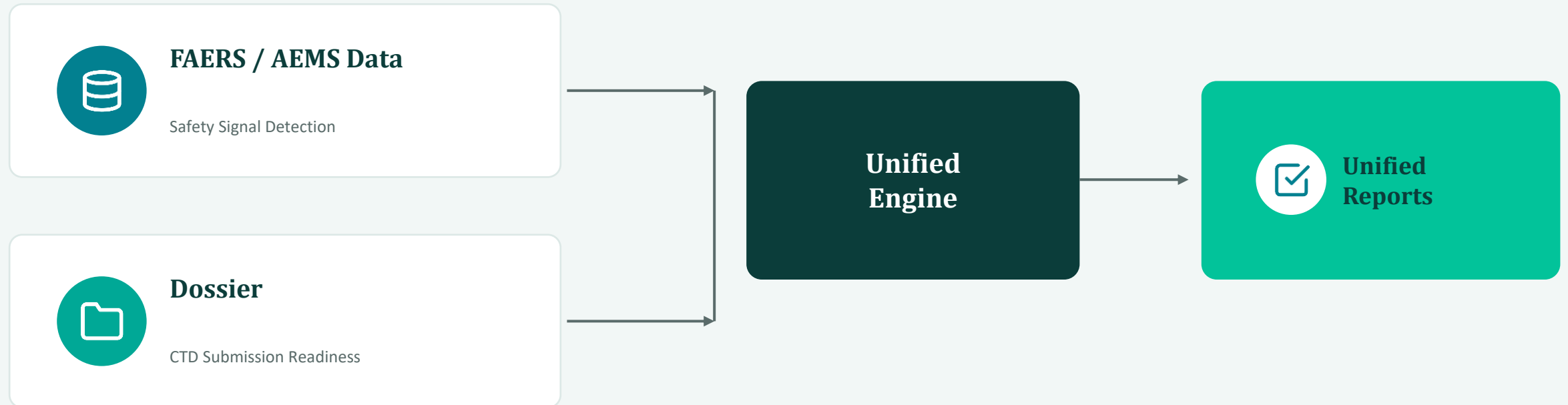


Disconnected workflows

Safety analysis and submission readiness are usually separate workflows.

One Platform, Two Engines, Unified Reports

The platform combines safety intelligence and regulatory readiness in a single workflow.



Two independent, deterministic engines feed one combined, judge-ready report.

What The Platform Actually Does



Safety Signal Detection

- Drug / event normalization
- PRR + Chi-square
- 2x2 contingency table
- Trend analysis
- Clustering
- Potential / Weak / No Signal classification



Submission Readiness

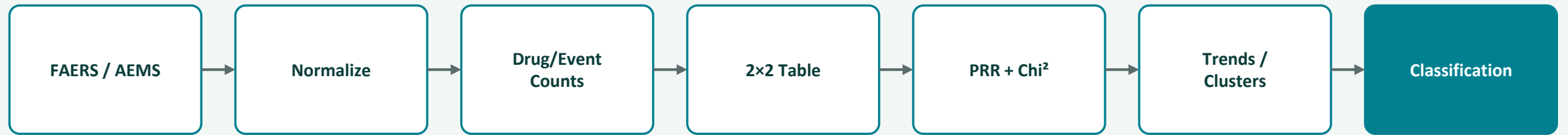
- CTD Modules 1–5
- Document mapping
- Present / Missing / Unrecognized / N-A
- Completeness scoring
- Gap report



Reports

- Safety report
- Readiness report
- Combined report
- PDF / CSV / JSON where implemented

Safety Signal Detection Pipeline



Potential Safety Signal

$\text{PRR} > 2$ AND $\text{Chi}^2 > 4$ AND
reports (a) ≥ 3

Weak / Emerging Signal

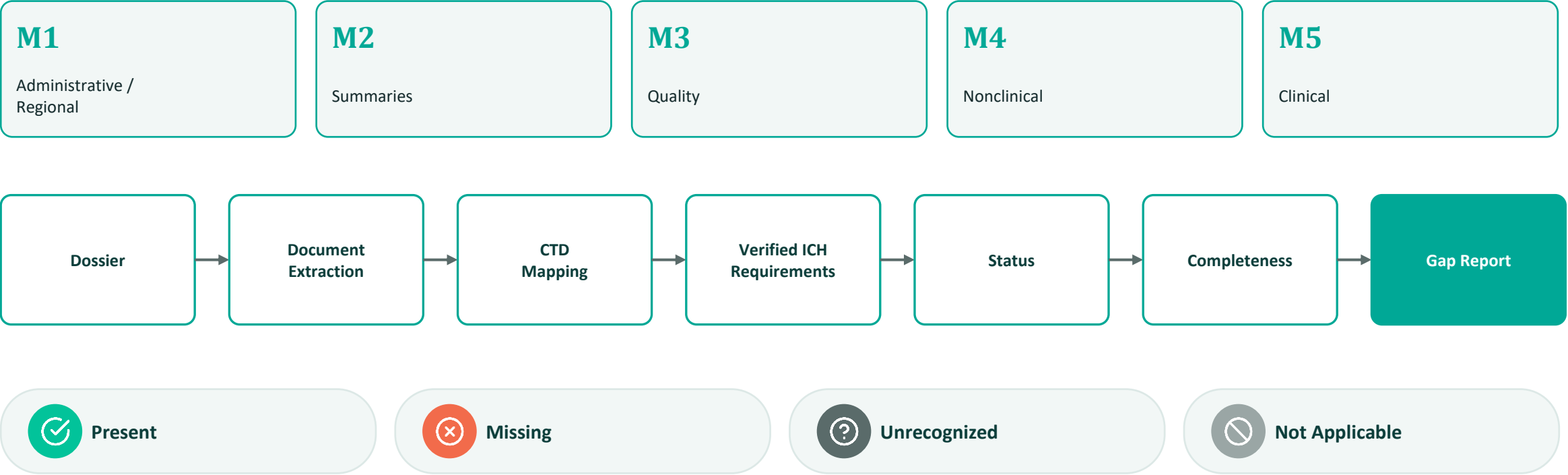
$\text{PRR} > 2$ AND $\text{Chi}^2 > 4$ AND
reports (a) < 3

No Signal

All other cases

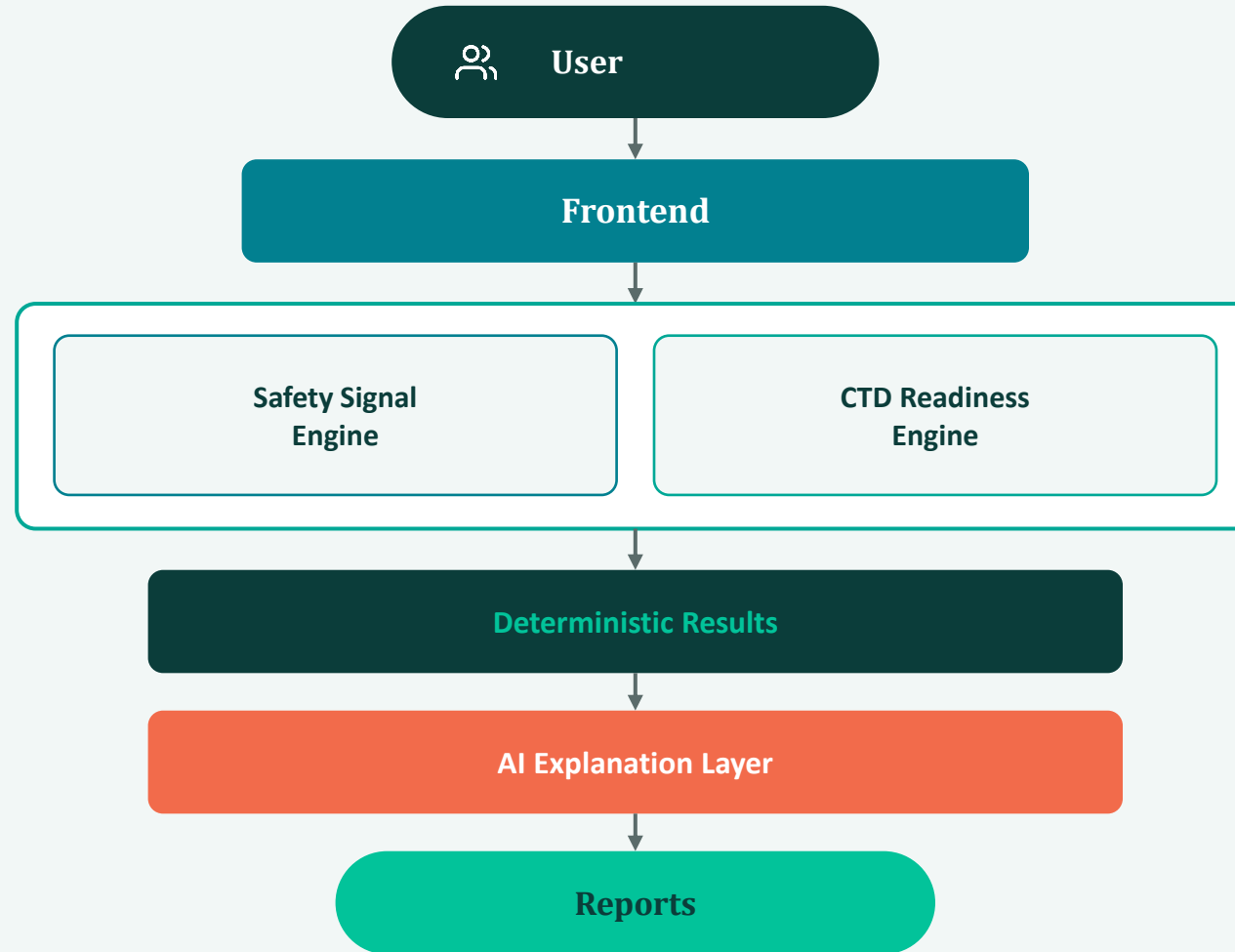
Statistical association is not proof of causality.

Submission Readiness Pipeline



AI does not invent ICH requirements or calculate readiness.

Deterministic Core, AI-Assisted Explanations



Statistical calculations are deterministic.



CTD completeness is deterministic.

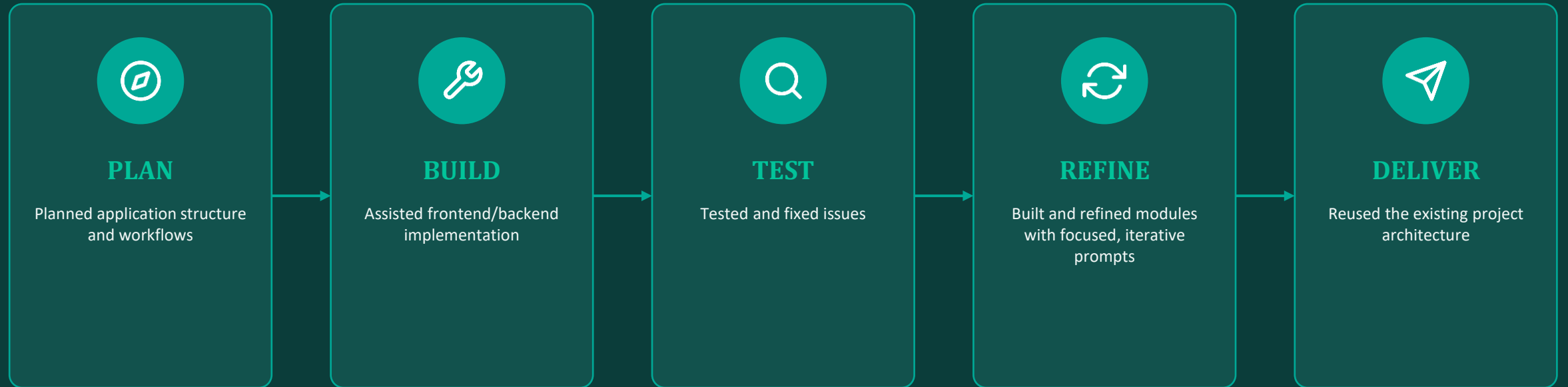


AI assists with explanations only.



Human experts remain responsible for final decisions.

AI-Accelerated, Human-Led Development



"IBM Bob accelerated development; technical decisions and validation remained human-led."

What We Know We Don't Know Yet

Limitations

- FAERS/AEMS reporting bias
- PRR does not prove causality
- Small-count sensitivity
- Data quality and duplicate reports
- CTD completeness does not guarantee approval
- Module 1 is region-specific
- Document mapping can be ambiguous
- AI explanations require human review

Future Scope

- EBGM / MGPS and advanced signal detection
- Better trend/cluster analysis
- More regulatory regions
- Improved document classification
- Regulatory requirement version tracking
- Human-in-the-loop AI assistant
- Additional safety data sources

From Signal to Submission-Ready



"From detecting potential safety signals to understanding regulatory readiness."

Thank You — Questions?